Fairness Has Memory

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Abstract

When a scarce, indivisible good is allocated by lottery among people with equal claims, the standard theory of fairness says the lottery is the fairest thing available. When the same good is allocated the same way year after year to largely the same people, the theory has nothing further to say: each lottery is fair, so the sequence is. This is a mistake. A sequence of individually fair lotteries can be an unfair allocation, and where claimants stay in the pool after being served, fairness requires that their chances in later rounds depend on the history of their unsatisfied claims. The first two claims use only premises the standard theory accepts: that a divisible good must be divided in proportion to claims, and that a lottery is a second-best surrogate for division. Over a sequence, the good becomes divisible. I then show what memory should track, units where later units are as good as earlier ones and waiting where they are not, and how heavily it may press: as far as the chance owed to those who may leave before the division reaches them allows, the two being weighed as Broome weighs one imperfect fairness against another. Where the good recurs that is far, because past winners' surplus pays for memory; where winners leave the pool and some will leave unserved, nobody present holds a surplus, the theory fixes only the frontier, and I argue for its forgetting end. Rules that forget and rules that remember by points or queues are both unfair where the good recurs, the second because a gentler memory beats them on every count; where winners leave the pool and some will leave unserved, the queue is forbidden, and I argue that the lottery that forgets is fairer than any weight on waiting.

1 Two lotteries

Every year Nevada and New Mexico have more people who want to hunt mule deer than tags to give, so each holds a lottery. Nevada remembers: an unsuccessful applicant earns a bonus point, her points are squared to fix how many random numbers are drawn for her the next year, and her lowest number is her entry, so ten years of losing gives her a hundred and one draws against a newcomer's one.[1] New Mexico forgets: "New Mexico does NOT grant preference to applicants who were unsuccessful in previous drawings."[2] A hunter passed over for ten years enters the eleventh draw as a first-time entrant.

Many institutions side with Nevada, though they disagree about how: Arizona adds one random number per unsuccessful year and reserves a fifth of its tags for the highest point-holders; Colorado gives most licences to whoever has the most preference points, chance only breaking ties. The Grand Canyon's river-permit lottery weights by years since an applicant's last trip, capped at five, and starts a newcomer at the cap.[3] Others forget, among them the lotteries that allocate some 85,000 H-1B visas and up to 55,000 Diversity Visas a year: a worker not selected in three successive years enters the fourth draw as a newcomer.

[1] Nevada Department of Wildlife, "Bonus Point Program," https://www.ndow.org/blog/bonus-point-program/.

[2] New Mexico Department of Game and Fish, "How the New Mexico Draw Works," https://wildlife.dgf.nm.gov/hunting/applications-and-draw-information/how-new-mexico-draw-works/.

[3] National Park Service, "Weighted Lottery," https://grcariverpermits.nps.gov/faq.cfm.

The theory of fair allocation has barely touched this disagreement, and where it does, sides with New Mexico. The best-developed account, John Broome's (1984a, 1991a, 1991b), holds that fairness requires claims to be satisfied in proportion to their strength, and that when the good is indivisible and claims are equal, an equal lottery is the fairest available procedure. It evaluates a lottery on its own, and on its own each year's New Mexico draw is impeccable. Broome's critics, who locate the fairness of lotteries in what they express or exclude rather than in the chances they distribute (Wasserman 1996; Stone 2007, 2011; Spiekermann 2022), reach the same verdict: an impartial procedure is fair whatever last year's did. The wider literature takes the single lottery as its unit (Sher 1980; Kornhauser and Sager 1988; Elster 1988; Duxbury 1999; Saunders 2008; Stone 2008). The nearest to my thesis is a page of Hersch and Rowe's (2025, 3): "many goods can be divided temporally as well as spatially," so that an indivisible good can be divided "over time through a rotation," which they credit to Elster (1989), and for recurring chores "who completed the chore last time matters to who has a claim to avoid it this time." That is the divisibility argument in miniature. In computational social choice (Lackner 2020; Lackner and Maly 2023; Freeman et al. 2017; Benade et al. 2018) history-dependence is an axiom. What follows adds the derivation inside Broome's theory, against his own extinction thesis; the constraint on how hard memory may press; and the structure in which it is all but forbidden.

I will argue for three claims. *Non-separability*: the fairness of a sequence of allocations is not a function of its members' fairness as the standard theory assesses it. *Memory*: when units of a good are allocated repeatedly among claimants who remain in the pool after receiving one, fairness requires that each claimant's chances in later rounds depend on her history of unsatisfied claims. *Currency and weight*: what memory tracks is a claimant's shortfall from her fair share, in units of the good where a later unit is as good to her as an earlier one and in waiting where it is not; and it may press that shortfall only as far as the chance owed to those who may leave before the division reaches them allows, the two weighed in Broome's own manner.

2 Claims, chances, and division

A claim, in Broome's sense, is a duty owed to the candidate herself that she should have the good, as distinct from other reasons for giving it to her, such as usefulness (Broome 1991a, 91–93). Fairness requires that "equal claims require equal satisfaction," that "stronger claims require more satisfaction than weaker ones," and that "weaker claims must not simply be overridden by stronger ones" (Broome 1991a, 95).

Two features of the theory carry my argument. The first I will call the *Division Principle*: where a good is divisible, proportional satisfaction is achievable and required: if two people have equal claims to some good, it "should be divided," even when giving it all to one of them would do more good (Broome 1991a, 95). Broome states it most generally in a passage that already divides across time:

> The good, for instance, of avoiding military service could be divided by conscripting everyone for a short time rather than a few people for a long time. If people's claims to a good are equal, then fairness requires that the good should be divided between them if it can be. That is the only way to treat them with perfect equality. (Broome 1984a, 47)

The second is the account of lotteries as second best. When the good is indivisible and scarce, a lottery gives "a sort of partial equality in satisfaction": "Each person can be given a sort of surrogate satisfaction. By holding a lottery, each can be given an equal chance of getting the good. This is not a perfect fairness, but it meets the requirement of fairness to some extent" (Broome 1991a, 97–98). Broome grants this only if a chance counts as surrogate satisfaction, which he finds plausible (98). Two things follow. Where division is available, it is required. Where it is not, the

surrogate is owed: a claimant who cannot be given her portion must be given her chance.

What duty does Nevada owe anyone to let her shoot a deer? Broome declines to say which reasons are claims, beyond granting that some are (Broome 1991a, 93). The claims in my cases arise from the allocator's undertaking: a body that announces it will allocate a public resource fairly among eligible applicants thereby owes each of them fair treatment, and what that requires is not the agency's to stipulate. New Mexico's announcement that it grants no preference to past losers describes its procedure, not its fairness.

3 The divisibility argument

Consider a good that comes in indivisible units, one per period, allocated by one agency among equal claims, where receiving a unit does not remove a claimant from the pool: a tag a year, a night off the rota a week. Suppose for now that the same N claimants are present for T periods. Within any one period the standard verdict is correct: one unit, N equal claims, no division, so an equal lottery. But the units are indivisible and the sequence is not. Over T periods it is possible to give each claimant T/N units, or as near as the integers allow. Call this *rotation*: a random order drawn once and cycled. Rotation is, in the sequence, what dividing the food is in the single case. The sequence can be divided, so fairness requires rotation, or whatever comes as close to it as circumstances allow.

Compare New Mexico's procedure, a fresh equal lottery every period: the *forgetting* rule. Over T periods it does not divide the sequence but raffles it. With ten claimants and ten periods it leaves, on average, three and a half with nothing while others receive two or more.[4] Rotation leaves nobody with nothing. The forgetting rule does over the sequence what the theory forbids in the single case: it takes a divisible good and gives it out by lot.

As a reductio: suppose someone proposed to allocate a cake among ten people with equal claims by cutting it into ten slices and raffling each slice separately. Each raffle, taken by itself, is what the theory recommends for an indivisible good. Nobody thinks this a fair way to allocate a cake; the theory itself says the cake should be divided, which means giving everyone a slice, not a chance at each. A cake handed out over ten days is still a divisible cake.

The first consequence is non-separability. The standard theory assesses a lottery by the claims present when it is held, fixed by need, desert and the like, not by earlier lotteries. Call a rule *locally fair* at a period if its procedure then, so assessed, is fair. The forgetting rule is locally fair at every period; rotation at none after the first, since given the history one claimant is certain of the unit. Yet rotation is the fair allocation of the sequence and the forgetting rule is not. So fairness, so assessed, is not separable across occasions. Put another way, as I will from §4 on: if claims carry history, so that a claimant who has received less than her share has a stronger claim to the next unit, then rotation *is* locally fair at every period and fairness is separable after all. These are one thesis in two vocabularies; what is not available is history-blind claims *and* separable fairness.

The second consequence is memory. Any rule that divides the sequence must give units to those who have received fewer, which requires knowing who they are.

The argument assumes something an opponent will press. The Division Principle applies to a good that "could be divided," and the tags a hunter draws in different years are not portions of one tag. What licenses treating them so is *Fungibility*: units allocated in different periods are portions of one divisible good, for a claimant, to the extent that her claim is to a unit of that kind and a later unit would satisfy it as an earlier one would. The hunter's claim is to a tag, not to the 2027 tag. Fungibility is a matter of degree, since delay is nearly always some cost, and it fails outright

[4] The expected number with nothing is $N(1-1/N)^T$. The simulation code behind every figure in this paper accompanies it.

when a later unit is much worse for the claimant: the visa applicant whose sponsor will not wait. Then the claim is largely to timeliness. But timeliness is divisible: delay can be shared out as units cannot, so the argument changes currency. Fairness requires that the *waiting* be divided in proportion to claims, where the constraint of §5 allows it (§5.1).[5]

4 The extinction objection

The argument seems to run into a doctrine Broome states in the same year. Once a fair lottery has given Michael the dialysis machine, it would be absurd to hold another on the ground that lotteries are fair: "once Michael has won the first lottery he has a perfect claim to the treatment, and Maggie has none. Consequently it is not at all fair to hold a second lottery" (Broome 1984b, 628–29). Call this the *extinction thesis*: losing a fair lottery extinguishes the loser's claim. Vong (2015) keeps the thesis in one form: a *procedural* claim, once satisfied by a fair lottery, is satisfied, while a *benefit* claim "cannot be lost due to the results of a procedurally fair lottery" (pp. 479–480); so when new units arrive the losers are "a new case of scarce benefit distribution" and get a fresh fair lottery (p. 483). In Vong's cases, where recipients leave the pool, that is a fresh lottery among those who did not benefit, which is the forgetting rule; and where the literature does give a lottery memory (Feldblyum Le Blevennec 2023), it is the *winner's* standing that is protected when new claimants arrive, and Broome himself, Vong reports (p. 481), suggested adjusting later lotteries' probabilities in the winner's favour so that every claimant has an appropriate chance of benefiting.

The reply distinguishes three structures the objection runs together. A *re-run*: the same claimants, the same unit, a second lottery because the first went the wrong way for someone. Broome is right that this is absurd, because there is nothing left to divide. An *interrupted allocation*, Vong's case: one unit, and new claimants arriving before it is handed over. And *recurrence*: a new unit each period, among overlapping claimants. Here the extinction thesis is true and irrelevant. Maggie's claim to the 2026 tag was extinguished when she lost; it is someone else's. But there is a 2027 tag, and the Division Principle says the two together are a divisible good, so the 2027 tag goes, or is more likely to go, to someone who did not get the 2026 one. It is a claim to a unit that exists, with more force than a past winner's for the same reason that, when a cake is divided, someone who has had no slice has more of her claim outstanding than someone who has had one. Where the currency is waiting and everyone will be served, the same holds with waiting in place of slices.

To bite, Broome's objection needs a stronger thesis: that losing a fair lottery for one unit extinguishes any bearing the loss might have on later units. That is what the cake refutes: someone who raffled each slice could say of each raffle that the losers' claims to that slice were fairly extinguished, and the cake would still not have been divided. Vong's form fails for a related reason: a procedural claim is satisfied by a fresh equal lottery only if a fresh equal lottery is the fair procedure, and over recurring units the Division Principle says it is not. Vong rejects views on which losing weakens a claim and on which winning strengthens it (pp. 477–479); extended to recurrence, his view that a benefit claim survives a loss is an ally of mine, and the question his objections, about new claimants for a single unit, do not reach is whether winning weakens a claim to *later* units.

A residual worry makes the surrogate character of chances do work. If the 2026 lottery gave Maggie surrogate satisfaction, does extra weight in 2027 not satisfy her twice? A chance is not a portion of the good but what one gives when a portion cannot be given. If an equal chance of a slice were a kind of equal slice, the raffle of the cake

[5] With unequal claims the problem is apportionment (Wintein and Heilmann 2018), whose divisor methods have a sequential form (Balinski and Young 2001, 100, Prop. 3.3). "Repeated" fair division (Igarashi et al. 2024; cf. Aleksandrov and Walsh 2020) concerns envy, not claims.

would be a division of the cake, and the theory could not say that the cake should be divided instead. What memory counts is units received, not chances.

5 What memory tracks

Sort the institutions by structure. In *recurrence* a claimant who receives a unit stays in the pool: tags, permits, chores. In a *bottleneck* she leaves: a visa, a flat, a kidney. (Hersch and Rowe use "bottleneck" more narrowly, for queues everyone eventually clears.) Structure is fixed by whether winners leave and currency by Fungibility, and in practice they run together, since whoever leaves on being served was waiting for one unit and cared when it came.

5.1 Shortfall in units

In recurrence the argument of §3 fixes what memory tracks: the difference between the share of units a claimant should have received by now and the number she has, her *shortfall*. With equal claims and one unit per period, a claimant present for r periods who has received w units has shortfall $r/N - w$. Shortfall is neither the number of losses nor time waited: points-for-losses systems reset on a win, treating a claimant who won once in twenty years as a newcomer, and count a loss among two hundred applicants as one among two; time waited tracks shortfall only if units and claimants are constant in number.

Consider the rules that press shortfall with a weight $\kappa \geq 0$: each claimant's chance is $1/N$ plus κ/N times the excess of her standing over the pool's mean, floored at zero, where standing is shortfall kept relative to the pool: a newcomer enters level with the pool's mean, and a claimant who leaves takes her standing with her (§6). At $\kappa = 0$ this is the forgetting rule. At $\kappa = 1$ a claimant's distance from the mean shrinks by about a factor of $1 - 1/N$ each period, so it is corrected over roughly the time the practice takes to hand out one round of shares. A past winner's chance falls below a newcomer's, and reaches zero once she is $1/\kappa$ units ahead of the pool; she is then excluded until the others catch up, but never taxed. As κ grows the rule approaches giving each unit to whoever is furthest behind, which is Aleksandrov et al.'s (2015) Balanced Like. In a fixed population everyone starts level, so each has the same chance of every unit whatever κ is, while dispersion falls: its variance is $\frac{N-1}{2\kappa N}$ absent clipping, and with ten claimants and ten periods the expected number left with nothing falls from 3.5 under the forgetting rule to 2.1 at $\kappa = 1$, 1.4 at $\kappa = 2$ and 0 at $\kappa = 10$. In a fixed population the Division Principle says: press as hard as you can, and rotate.

The rule has a near relative. Lackner's (2020, 2106) Perpetual Quota weights each voter by $\max(0, \text{quota} - \text{satisfaction})$, which in my setting is shortfall; but his rules are deterministic, so here it *is* rotation in its tie-breaking order, an arbitrary fixed order among equal claims, which the theory of claims forbids.

5.2 Newcomers and the surrogate constraint

Once claimants come and go, an objection appears. A newcomer has as much of her claim outstanding as any past loser had on first application, and may be gone before the division reaches her; under any rule with memory she is given a smaller chance than that loser now has.

The answer also fixes how heavily memory may press. Division of the sequence is available only to a claimant who stays long enough for it to be completed; one who leaves after three periods cannot be given three-tenths of a tag; for her the sequence is as indivisible as the unit. If the allocator knew who would leave, it would owe them the surrogate outright, a chance each period proportional to their claim, since a whole unit is more than their share. Not knowing, it owes the surrogate to everyone, in the only form available: that no claimant's expected share, over

whatever tenure she turns out to have, depart from proportionality. Call this the *surrogate constraint*. It is measured over the chances the rule assigns, exit taken as given; a claimant who leaves because she lost has a complaint against her luck, not the rule. The chance is what a claimant who will not stay for the division is *owed*; the dispersion among those who stay is what the Division Principle measures. Where neither departure from proportional satisfaction can be reduced without increasing the other they must be weighed. That is how Broome himself treats the fairness a lottery contributes and the less fair result it is likely to produce: the likelihood of the one "will have to be weighed against" the other (1991a, 99). The third claim uses that method and nothing stronger.

Three things hold of this constraint when exit is independent of history. First, it can only be met exactly: chances sum to one in every period, so under any rule the pool's expected share is proportional, and no tenure class can stand above proportionality unless another stands below it. Memory may redistribute realized shortfall across tenures, never expected share. Second, the shortfall rule meets it, for every κ, so long as no chance is clipped at zero. A newcomer enters level with the pool, so her chance in her first period is $1/N$ whatever the pool's state; thereafter her expected distance from the mean shrinks each period by the factor $1 - \kappa/N$ and gains no constant term, so it stays zero, and every tenure class receives exactly its share. Third, what memory can do has a closed form. With exit rate q the variance of shortfall at exit is about $\frac{N-1}{N(Nq+2\kappa(1-q))}$, so the reduction relative to forgetting depends on κ and on the number of units a claimant can expect to receive during a stay, $1/Nq$. Where a claimant can expect one unit during her stay, as in Table 1, $\kappa = 1$ removes over a third of the dispersion; where a hundred claimants compete for each unit and stay ten years, under a tenth; as units per stay vanish, the verdict of this section shades into that of §5.1.

Clipping breaks the exactness, and no weight escapes it: at any $\kappa > 0$ a claimant far enough ahead of the pool wins with positive probability, her chance cannot fall below zero, and her unpaid share then falls on everyone, the newcomer included. So no rule with memory meets the constraint exactly under churn, and the weighing cannot be avoided. What each rule costs the leavers and buys the stayers, Table 1 shows.

Table 1: Ten claimants with equal claims, one unit per period, each leaving with probability 0.1 per period and replaced; 200 runs of 1,500 periods. A newcomer's chance in her first period and expected shares at longer tenures, as multiples of the proportional $1/N$ per period; standard errors at most .003 in the first column and .02 elsewhere. Dispersion is the standard deviation, in units, of a claimant's shortfall when she leaves.

Rule	Newcomer's chance	Expected share, tenure 10	Tenure 30	Dispersion of realized shortfall
Forgetting ($\kappa = 0$)	1.000	0.99	0.97	0.95
Shortfall, $\kappa = 1/2$	1.000	0.99	1.02	0.71
Shortfall, $\kappa = 1$	0.993	1.00	1.00	0.60
Shortfall, $\kappa = 2$	0.943	1.00	1.01	0.51
Shortfall, $\kappa = 10$	0.523	0.97	0.99	0.42
Points for losses, linear (Arizona, random pass)	0.263	0.96	1.15	0.69
Points for losses, squared (Nevada)	0.082	0.99	1.21	0.58

Rule	Newcomer's chance	Expected share, tenure 10	Tenure 30	Dispersion of realized shortfall
Queue / preference points (Colorado)	0.000	1.01	1.26	0.47
Years since last win, capped (Grand Canyon)	1.169	1.00	0.99	0.80

The exchange is lopsided. The leak costs the newcomer three hundredths of one per cent of a period's chance at $\kappa = 1/2$ and under one per cent at $\kappa = 1$, against reductions in dispersion of a quarter and over a third: in units of the good, a hundredth of a unit over a ten-period stay against a third of one. No weighing that counts both can prefer the forgetting rule. At $\kappa = 2$ and $\kappa = 10$ the leak is six per cent and nearly half, and the exchange becomes a real one.

In Table 1's setting the rules in use are not on that exchange at all: each is dominated. Linear points give a newcomer a quarter of the chance her claim warrants and a claimant who stays thirty periods 15 per cent more than her share, with more dispersion than $\kappa = 1$; squared points cut the newcomer to a twelfth, with more dispersion than $\kappa = 2$; a queue gives her nothing, and whoever leaves early leaves with nothing, with more dispersion than $\kappa = 10$; the Grand Canyon's rule gives her a sixth more than her share, which by conservation the others pay for, with more dispersion than $\kappa = 1/2$. For each there is a shortfall rule that does better by the newcomer and on dispersion at once, so each is forbidden whatever weight those two are given.

Within the family the weight is a genuine choice, and the theory fixes the frontier rather than the point on it. One point is salient in its own currency. A claimant's chance reaches zero when she is $1/\kappa$ units ahead of the pool, and the Division Principle counts in units; at $\kappa = 1$ exclusion begins exactly when she holds a whole unit that is someone else's share, a larger κ excludes for fractions of the indivisible unit, and a smaller one leaves a chance with someone who already holds another's unit. I adopt it.

So, in recurrence, the third claim is established. Memory is required: the forgetting rule gives up a third of the division for a leak too small to weigh. To remember by points or by queue is forbidden: in Table 1's setting each such rule is beaten on both counts by one that remembers less harshly, and where units per stay are few, so that the queue can join the frontier, it is excluded by the argument of §5.1 instead. At $\kappa = 1$ every tenure class stays within one per cent of its share; the residual lottery is the surrogate the short-stayers are owed. Only where the pool is fixed is it zero, and there the rule is rotation.

5.3 Bottlenecks and waiting

In a bottleneck nobody accumulates more than one unit; what varies is time waited. Fungibility fails, because each claim is to one unit and to having it soon; what is divisible is the waiting. Under the forgetting rule, waiting is raffled: with a pool of ten kept full by arrivals and one unit per period, about one in seventy waits forty periods or more, though everyone's expected wait is nine; a queue serves everyone at nine. Time, unlike a flat, is divisible, so the Division Principle applies to the waiting and says: equalize it. This is why John and Millum (2020, 181, 205–6) are right that a queue can be as fair as a lottery, and Hersch and Rowe (2024) right to separate "bottleneck" cases, where queues belong, from "scarcity" cases, where lotteries do; between them lies recurrence, where a queue gives a short-stayer nothing.

The surrogate constraint carries over, and here it cuts the other way. Where claimants may leave unserved, by dying

or losing eligibility, and the allocator cannot tell who, those to whom the division is not available are owed the surrogate: an equal chance, each period, at the unit. In recurrence the leavers paid almost nothing for memory because the stayers' own past wins paid for it. In a bottleneck nobody present has received a unit, so there is no surplus to draw on: the shortfall rule itself reduces to forgetting, and every unit of chance a rule gives to a long waiter is taken from someone whose claim is wholly outstanding. The frontier is correspondingly steep. With a pool of ten, one unit a period and one claimant in a hundred leaving unserved each period, a queue gives a newcomer no chance and the served a spread of waits under a period; a mild weight on waiting gives her three-fifths of her chance and the served a spread of six and a half periods; forgetting gives her the whole of it and a spread of nearly nine. The fraction who leave unserved is the same under all three, since chances sum to one; what the rules choose is who they are. The weighing is a different one too. What a weight on waiting buys is equal waiting among people who will all be served; what it costs is the chance of the good itself to those who will not be, for whom the chance is all fairness can give; and the rate between a unit and a period of waiting is the degree to which Fungibility fails, which the theory leaves to the case. Two things it does fix. The queue is forbidden wherever exit is unforeseeable: it gives whoever leaves early no chance at all, so that her claim is overridden rather than given the "some satisfaction" Broome requires of every claim (95, 98–99); it is "never on the cards" (98). The theory forbids the exact zero; how far above it fairness requires is the weighing that follows. And between forgetting and a weight the theory fixes the frontier, not the point; I judge forgetting the fairer, since what the unserved lose is the good and what the served gain is its timing, and say so; and where Fungibility fails hardest, as for the visa applicant, those who leave unserved are the very claimants for whom timing was the claim, so the chance now is what they are owed. So the result is two-sided a second way. Where everyone will be served, the waiting is divisible and fairness requires a queue; where some will leave unserved and the allocator cannot tell who, the queue is forbidden and memory is dearly bought. Memory is required in one structure and all but forbidden in the other; the difference is whether there are past winners whose surplus can pay for it. Housing lists and transplant lists, where some always leave unserved, fall on the second side so far as exit is unforeseeable. Where it is foreseeable, or rises with waiting, the claim is urgent, time waited is evidence of urgency, and the rule must weight it, as transplant allocation does: a difference in claims, not a memory of history.

The verdict for the visa lotteries follows, and it is not what the opening suggests. They are bottlenecks with attrition: a selected worker leaves the pool, and an unselected one may lose her sponsor or status unforeseeably. So the lottery that forgets is, by that judgement, the fairer rule, and Hersch and Rowe's (2024, 188) endorsement of the Diversity Visa lottery is right, though for a different reason: the surrogate owed to those who will leave unserved cannot be paid by anyone else. And their remark that a periodic lottery in a bottleneck is unfair because a person who "arrives into the lottery pool early and is continuously unlucky" may wait far longer than a lucky newcomer (199) holds only where everyone will be served.

6 How far does memory reach?

If Nevada must remember Maggie's 2026 loss in 2027, must it remember her loss in the housing lottery? In Arizona's draw? A view that cannot stop somewhere principled collapses into a demand that luck be equalized, a more contested thesis. Two bounds, besides the bound in weight from §5, follow from the argument that generates the claim.

Memory is bound to the practice. The corrective claim arises because a particular agency's allocation of a particular good left residual unfairness: a claim on *those* units, addressed to *that* agency. The good matters because the 2027

tag can divide the sequence only because it is a tag; a flat is no portion of what the hunter has a claim to. Goods are individuated by claims: a Nevada tag and an Arizona tag are different goods, two draws by one agency for the same tag one good, so an allocator cannot shed its obligation by splitting a practice. The agency matters because another allocator has none of its units to divide. This distinguishes the thesis from luck egalitarianism (Dworkin 1981; Cohen 1989; Temkin 1993), whose currency is compensation, so that a shortfall in one good can be made up in another; mine is silent on whether the loser is worse off.

Memory is bound to the claim. The corrective claim is a strengthened claim to the good, not a credit. Someone who applied unsuccessfully for ten years and then gave up hunting is owed nothing; there is no tag she has a claim to. Memory is not compensation: shortfall arises under the fairest rule, since the residual lottery produces it, so correcting it presupposes no wrong.

7 Objections

Losing on purpose. That memory rewards losing is a familiar objection: the NBA moved from a worst-first draft order to a lottery because of the incentive to lose (Elster 1988, 123 n. 46). In my cases losing is not under the claimant's control: she can only apply, and applying without a genuine claim is the problem of verifying claims, which every allocation faces. Where losing is controllable, as in a league, fairness is in tension with incentive, which is a reason to weigh it against other things, not evidence that it requires forgetting.

Equality, not fairness. Someone might grant the verdict and deny that fairness is what condemns the outcomes: what is bad is inequality, or that some are badly off (Parfit 2000). Suppose the persistent loser is, all things considered, no worse off than the repeat winner. Her complaint survives: this agency, dividing this good among equal claims, has given her less than her share. A priority view cannot explain why that complaint lapses when she stops claiming.

Ex ante. The last objection comes from the view, descending from Diamond's (1967) reply to Harsanyi (1955) (Frick 2015; Otsuka and Voorhoeve 2009), that a procedure's fairness is a matter of the prospects it gives people at the time of choice. On that view the forgetting rule and rotation are equally fair, since each claimant starts with the same prospects under both; the forgetting rule's defect, as the first row of Table 1 showed, is entirely ex post. First reply: the ex ante view must say when the ex ante is. At the beginning of the practice, forgetting and rotation are identical, and the Division Principle settles what the view leaves open; at the beginning of each period, the view asserts separability with respect to history-blind claims, which the cake refutes. Second, my view is ex ante exactly where the ex ante view is right: a claimant is owed chances insofar as division is unavailable to her and units insofar as it is, which is why the surrogate constraint conditions on her tenure, which settles whether division was available to her, and not on her draws, which are the surrogate itself. Dated at her entry, every rule gives her the same prospects, by conservation; the ex ante view, built for one-shot cases where division never is, can tell none of them apart.

8 Conclusion

Fairness has memory where the good recurs. What the memory tracks is a claimant's shortfall from her share of the good being divided; it is owed by the agency dividing the good, for as long as she has a claim to it; and it may be pressed only as far as the chance owed to those who may leave allows, which where the good recurs is far, since past winners pay. Where winners leave and some will leave unserved, nobody present holds a surplus, every weight on waiting is paid by those who will never be served, and I have argued that the weighing tells against memory. The

result was visible in Broome's own illustration of the Division Principle: military service, divided by conscripting everyone for a short time. Everyone's short time is a rotation. The lottery is what one does when one cannot rotate, and one can rotate more often than its defenders have noticed.

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